BIOINFORMATICS

WEEK 1 – DAY 2

DATA

- Data is distinct pieces of information, usually formatted in a special way.
- Information - When data is processed, organized structured or presented in a given context so as to make it useful it is called information.

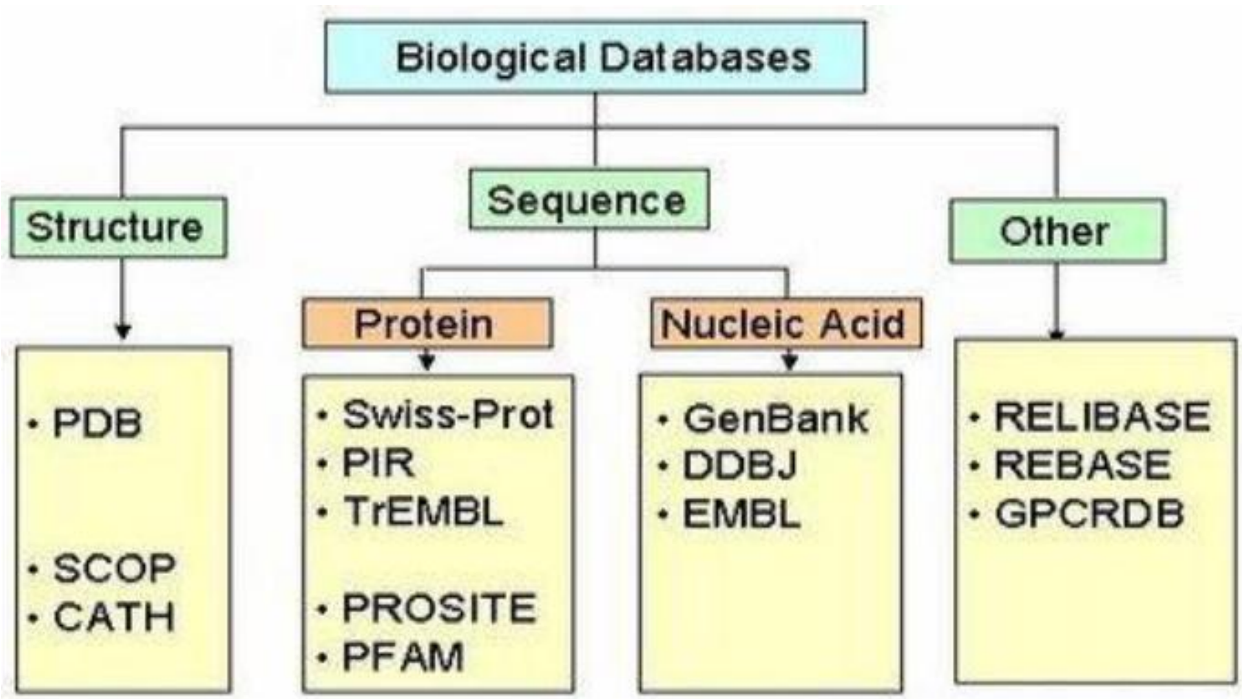
DATABASE - A database is an organized collection of data, which is accessible in various ways.

BIOLOGICAL DATABASES

- A biological database is a large, organized body of persistent data, usually associated with computerized software designed to update, query, and retrieve components of the data stored within the system.
- Biological databases are libraries of life sciences information, collected from scientific experiments, published literature, high-throughput and computational analysis.

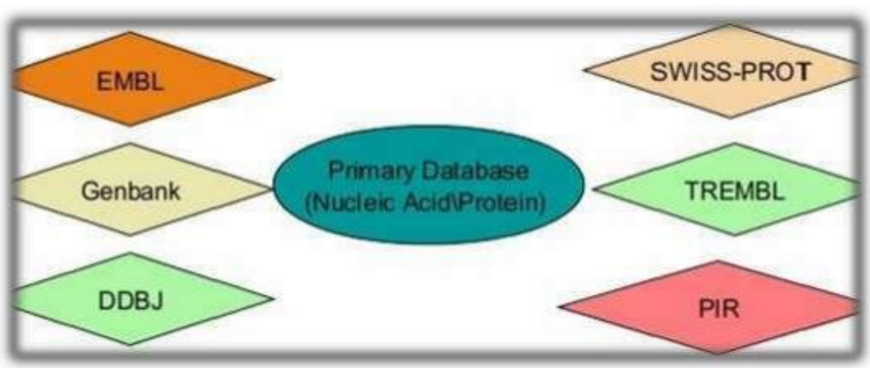
TYPES OF BIOLOGICAL DATABASES

- Primary Databases
- Secondary Databases
- Other Specialized Databases



PRIMARY DATABASES

- Primary databases contains biological data in its original form.
- Experimental results are submitted directly into the appropriate database by researchers.
- Once a database ACCESSION NO. is given to an kind of data in primary database then that data can't be changed further.



PRIMARY DATABASES – GEN BANK

- Gen bank is a database from NCBI , includes sequences from publicly available resources.
- GenBank is the genetic sequence database , an annotated collection of all publicly available DNA sequences.

- GenBank is part of the International Nucleotide Sequence Database Collaboration, which comprises the DNA Databank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI. These three organizations exchange data on a daily basis.
- https://www.ncbi.nlm.nih.gov/genbank/

PRIMARY DATABASES – EMBL • EUROPEAN MOLECULAR BIOLOGICAL LABORATORY

- It is an Nucleic acid Database that comes under EBI (European Bioinformatics Institute).
- It was Established in collaboration with DDBJ and GenBank.
- EBI's Sequence Retrieval system (SRS) is a network browser for databanks in molecular biology, integrating and linking the main nucleotide and protein databases.
- https://www.embl.org/

PRIMARY DATABASES – DDBJ

- The DNA Data Bank of Japan is a biological database that collects DNA Sequences.
- DDBJ is located at the National Institute of Genetics in the Shizuoka Prefecture of Japan.
- It is also a member of the International Nucleotide Sequence Database Collaboration.
- https://www.ddbj.nig.ac.jp

PRIMARY DATABASES – SWISS PROT

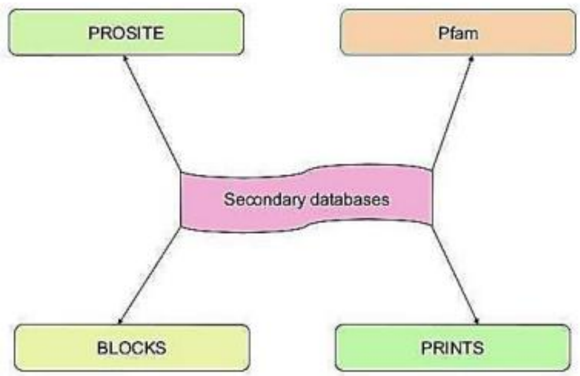
- SWISS – PROT was created in 1986 by Amos Bairoch with Swiss Institute of Bioinformatics.
- SWISS – PROT is a curated protein sequence database which strives to provide a high level of annotation.
- https://www.uniprot.org/

PRIMARY DATABASES – PIR

- Protein Information Resources (PIR) was established in 1984 by the National Biomedical Research Foundation (NBRF) as a resource to assist researchers in the identification and interpretation of protein sequence information.
- The PIR is an integrated public resource of protein informatics that supports genomic and proteomic research and scientific discovery.
- PIR maintains the Protein Sequence Database (PSD), an annotated protein database containing more than 283000 sequences covering the entire taxonomic range.
- https://proteininformationresource.org/

SECONDARY DATABASES

- Secondary databases contains Data derived from results of analyzed primary data.
- The data in secondary database is either manually created or generated automatically.
- It contains some valuable information such as about mutations or evolutionary relationship.



SECONDARY DATABASES - PROSITE • PROSITE IS A PROTEIN DATABASE.

- It consist of entries describing the protein families , domains and functional sites as well as amino acid patterns and profiles in them.
- They are manually curated by a team of the Swiss Institute of Bioinformatics and tightly integrated into Swiss prot protein annotation.
- https://prosite.expasy.org/

SECONDARY DATABASES - PFAM

- Pfam is a database of protein families that includes their annotations and multiple alignments generated using hidden markov models.
- The general purpose of the pfam database is to provide a complete and accurate classification of protein families and domains.
- The Pfam website allows users to submit protein or DNA sequences to search for matches to families in the database.
- https://pfam.xfam.org/

STRUCTURAL DATABASES

- Biological databases can be divided into two main types which include primary databases and secondary databases.
- The Protein Data Bank maintained at Brookhaven is the world repository of experimentally determined protein structures.
- They all contains information generated from X- Crystallography and NMR Experiments.

STRUCTURAL DATABASES - PDB

- The Protein Data Bank (PDB) is a database for the three-dimensional structural data of large biological molecules, such as proteins and nucleic acids.
- Comprises of
    - PDBe (PDB of Europe)
    - PDBj (PDB of Japan)
- https://www.rcsb.org/

STRUCTURAL DATABASES - SCOP • THE STRUCTURAL CLASSIFICATION OF PROTEINS (SCOP)

- database is a largely manual classification of protein structural domains based on similarities of their structures and amino acid sequences.
- http://scop.mrc-lmb.cam.ac.uk/

STRUCTURAL DATABASES - CATH

- The CATH Protein Structure Classification database is a free, publicly available online resource that provides information on the evolutionary relationships of protein domains.
- The four main levels of the CATH hierarchy:
    - Class
    - Architecture
    - Topology/fold
    - Homologous superfamily
- http://cathdb.info/

METABOLIC PATHWAYS DATABASES

- SMPDB Small Molecule Pathway Database - https://smpdb.ca/
- KEGG - Kyoto Encyclopedia of Genes and Genomes - https://www.genome.jp/kegg/

PYTHON

- Python is a high-level, interpreted, general-purpose programming language known for its clear syntax and readability.
- Created by Guido van Rossum and first released in 1991,

VARIABLES - A variable is simply a named container that stores information.

Think of it as a labeled test tube.

Test Tube Label → Gene

Inside → INS

In Python: gene = "INS"

Here:

- gene is the variable.
- "INS" is the value stored inside it.

PRACTICE

1. Create a new file: day2_variables.py

Type:

```
gene = "INS"

print(gene)
```

Output: INS

2. Multiple Variables

```
gene = "INS"

organism = "Homo sapiens"

protein = "Insulin"

print(gene)

print(organism)

print(protein)
```

Output

INS

Homo sapiens

Insulin

CORE DATA TYPES

- **str (String)**: Text data wrapped in quotes.
  name = "Alice"

- **int (Integer)**: Whole numbers without decimals.
  age = 25

- **float (Floating point)**: Numbers with decimal points.
  price = 19.99

- **bool (Boolean)**: Logical values that are either true or false.
  is_logged_in = True

COLLECTION DATA TYPES

- **list (List)**: Ordered, changeable collections of items.
  fruits = ["apple", "banana", "cherry"]

- **dict (Dictionary)**: Key-value pairs for mapping data.
  user = {"username": "coder123", "score": 95}

Strings - DNA sequences are just strings.

Example: dna = "ATGCGTTACG"

[Python treats DNA exactly like text.]

Print the DNA

```
dna = "ATGCGTTACG"

print(dna)
```

Length of DNA

```
dna = "ATGCGTTACG"

print(len(dna))
```

Output : 10

Comments - Comments explain your code. Python ignores them.

```
# This is the insulin gene

gene = "INS"
```

Good programmers comment their code.

Input - Instead of writing everything yourself, ask the user.

```
gene = input("Enter gene name: ")

print(gene)
```

Example = Enter gene name: TP53

Output : TP53

Combining Text

```
gene = "INS"

print("Gene:", gene)
```

Output - Gene: INS

FIRST BIOINFORMATICS PROGRAM

```
gene = input("Gene name: ")

organism = input("Organism: ")

print()

print("Gene:", gene)

print("Organism:", organism)
```

dna sequence length calculator

```
dna = input("Enter DNA sequence: ")

print("Length:", len(dna))
```

Example : ATGCGATCGATCG

Output - Length: 13

NUCLEOTIDE COUNTING

```
dna = "ATGCGATCGAA"

print("A =", dna.count("A"))

print("T =", dna.count("T"))

print("G =", dna.count("G"))

print("C =", dna.count("C"))
```

Output : A = 4

T = 2

G = 3

C = 2

GC CONTENT - the percentage of G and C bases in a DNA sequence. It's important because it influences DNA stability, primer design, and many sequencing applications.

```
dna = "ATGCGATCGAA"

g = dna.count("G")

c = dna.count("C")

gc = g + c

length = len(dna)

gc_percent = (gc / length) * 100

print("GC Content =", gc_percent)
```

Reading a FASTA File

Create a text file named: insulin.fasta

Paste: >Human insulin

ATGCCCTGTGGATGCGCTTCCTGCCCCTG

Now create: read_fasta.py

Write:

```
file = open("insulin.fasta")

content = file.read()

print(content)

file.close()
```

You just read a biological sequence from a file.

BIOPYTHON is a collection of Python tools made specifically for biology.

Instead of writing everything yourself, Biopython already knows how to:

- Read FASTA files
- Download sequences from NCBI
- Translate DNA to protein
- Perform sequence analysis
- Work with GenBank files

Open the terminal: pip install biopython